Guy's and St Thomas' NHS Foundation Trust

Royal Brompton & Harefield NHS Foundation Trust

Lung Volumes, Dead Space & Pulmonary Pressures



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With thanks to Isobel Davies

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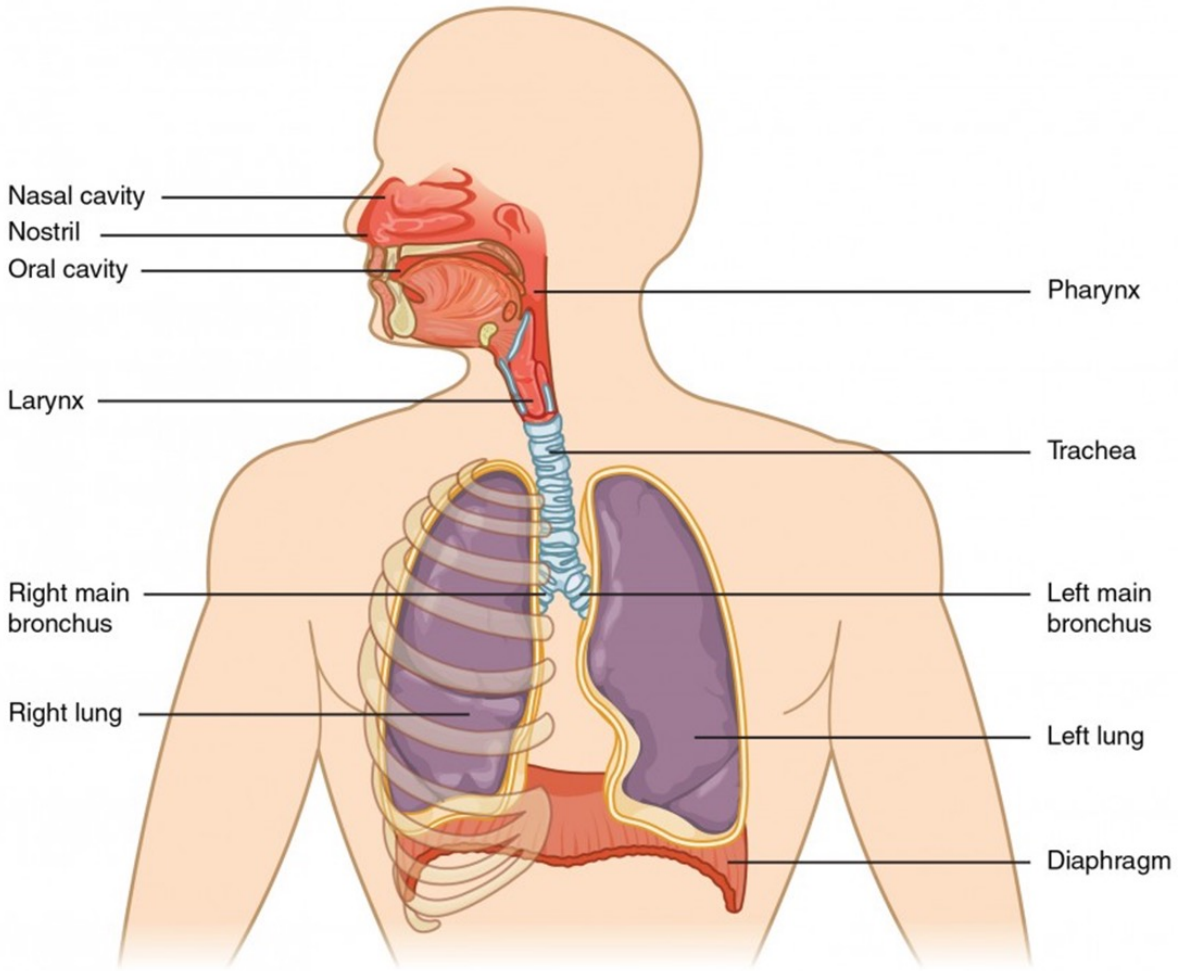
Learning Objectives:

- Describe the organisation and main structural features of the airways and alveoli
- Explain what is meant by respiratory dead space and its subdivisions: anatomical, alveolar and physiological
- List the major lung volumes and outline their measurement using spirometry and helium dilation
- Define the terms minute ventilation and alveolar ventilation
- Outline how intra pleural pressure, alveolar pressure, airflow change during quiet breathing, deep breathing and forced expiration
- Explain how intrapleural pressure can be assessed from oesophageal balloon pressure

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Primary functions of the lung

- Gas exchange
- Homeostatic regulation of body pH
- Protection from inhaled pathogens
- Vocalisation



Upper Respiratory Tract:

= ANATOMICAL DEAD SPACE

Made up of:

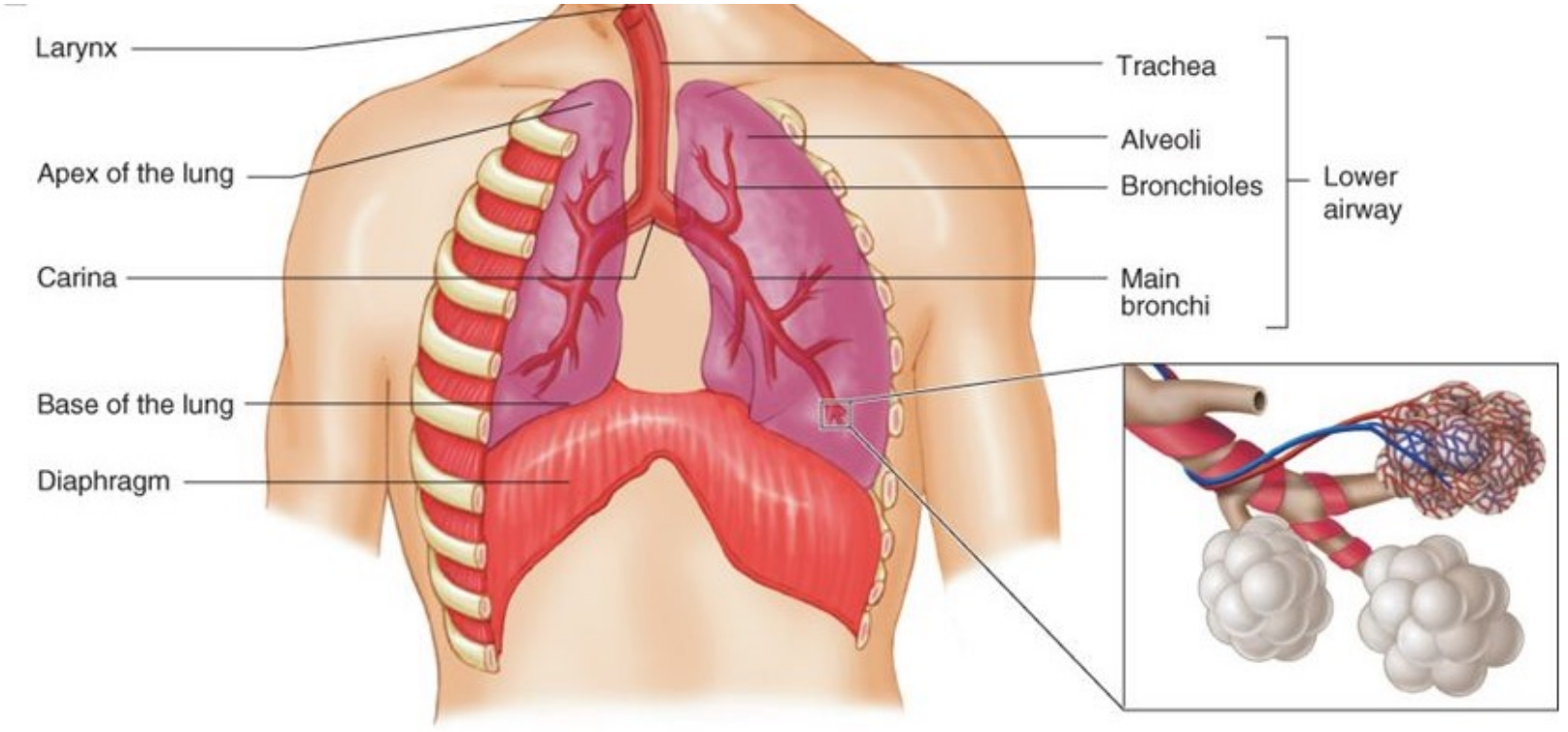
Nose

Nasal cavity

Pharynx

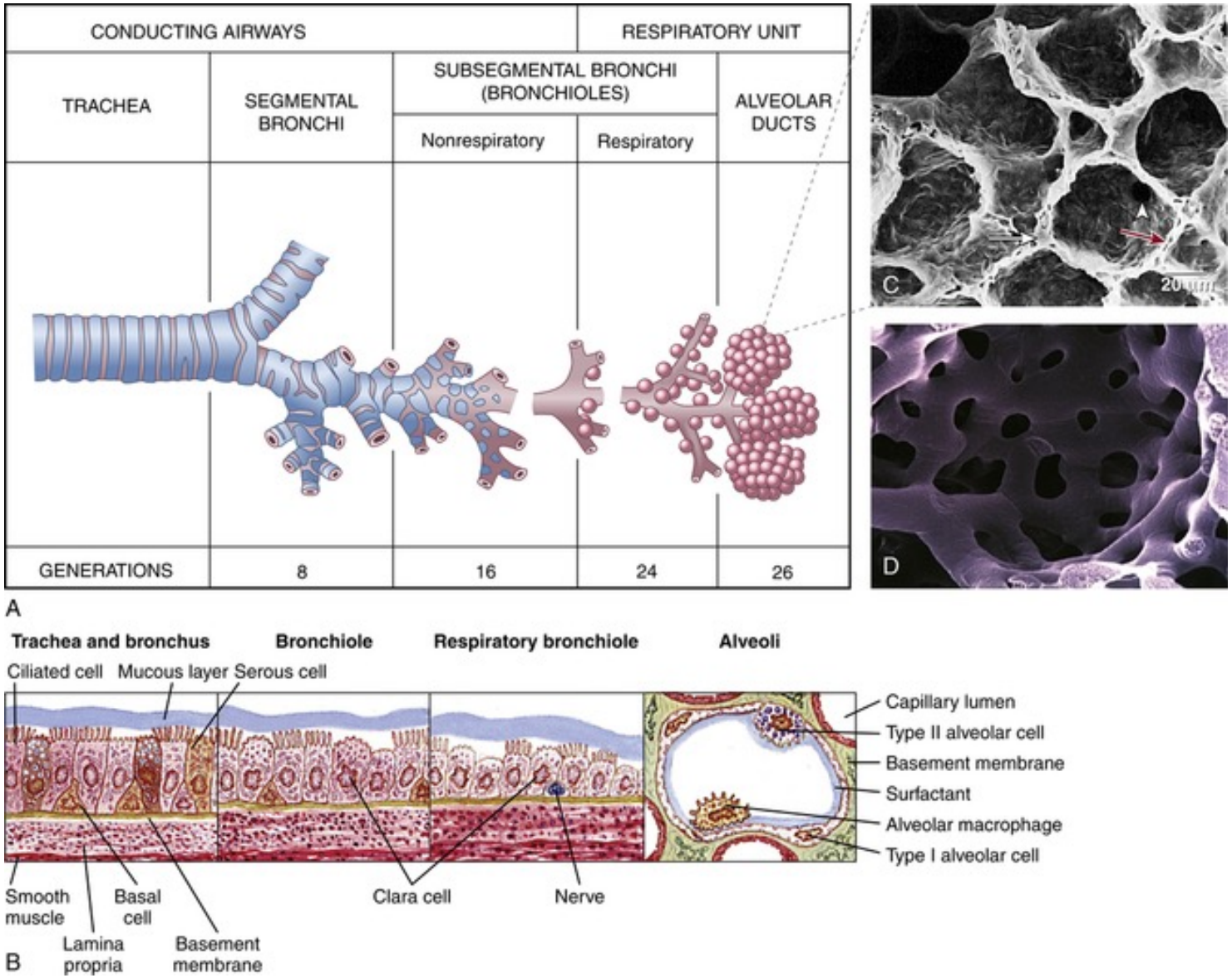
Larynx

LOWER RESPIRATORY TRACT:



Which aspects contribute to ‘dead space’, which are ‘respiratory zone’?

What does ‘total dead space’ or ‘physiologic dead space’ mean?


CONDUCTING AIRWAYS
RESPIRATORY UNIT
TRACHEA
SEGMENTAL BRONCHI
SUBSEGMENTAL BRONCHI (BRONCHIOLES)
Nonrespiratory
Respiratory
ALVEOLAR DUCTS
GENERATIONS
8
16
24
26
A
C
D
Trachea and bronchus
Bronchiole
Respiratory bronchiole
Alveoli
Ciliated cell
Mucous layer
Serous cell
Capillary lumen
Type II alveolar cell
Basement membrane
Surfactant
Alveolar macrophage
Type I alveolar cell
Smooth muscle
Basal cell
Lamina propria
Basement membrane
Clara cell
Nerve
B

‘Dead Space’

Volume of air that is inhaled & does not take part in gas exchange

Physiological dead space = anatomical dead space +alveolar dead space

Anatomical:

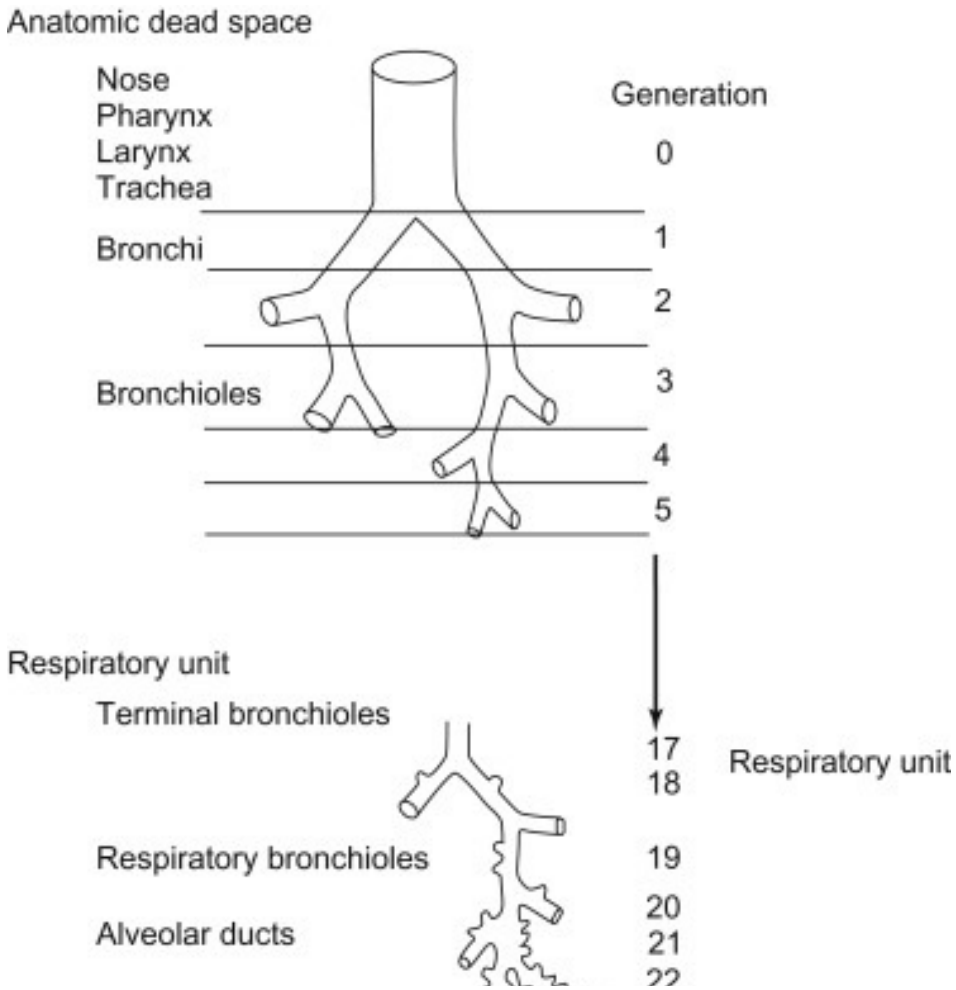
Volume of the conducting airways

Measured using Fowler’s Method (nitrogen washout technique)

Normal values can be estimated as a 1/3 of resting tidal volume

Alveolar:

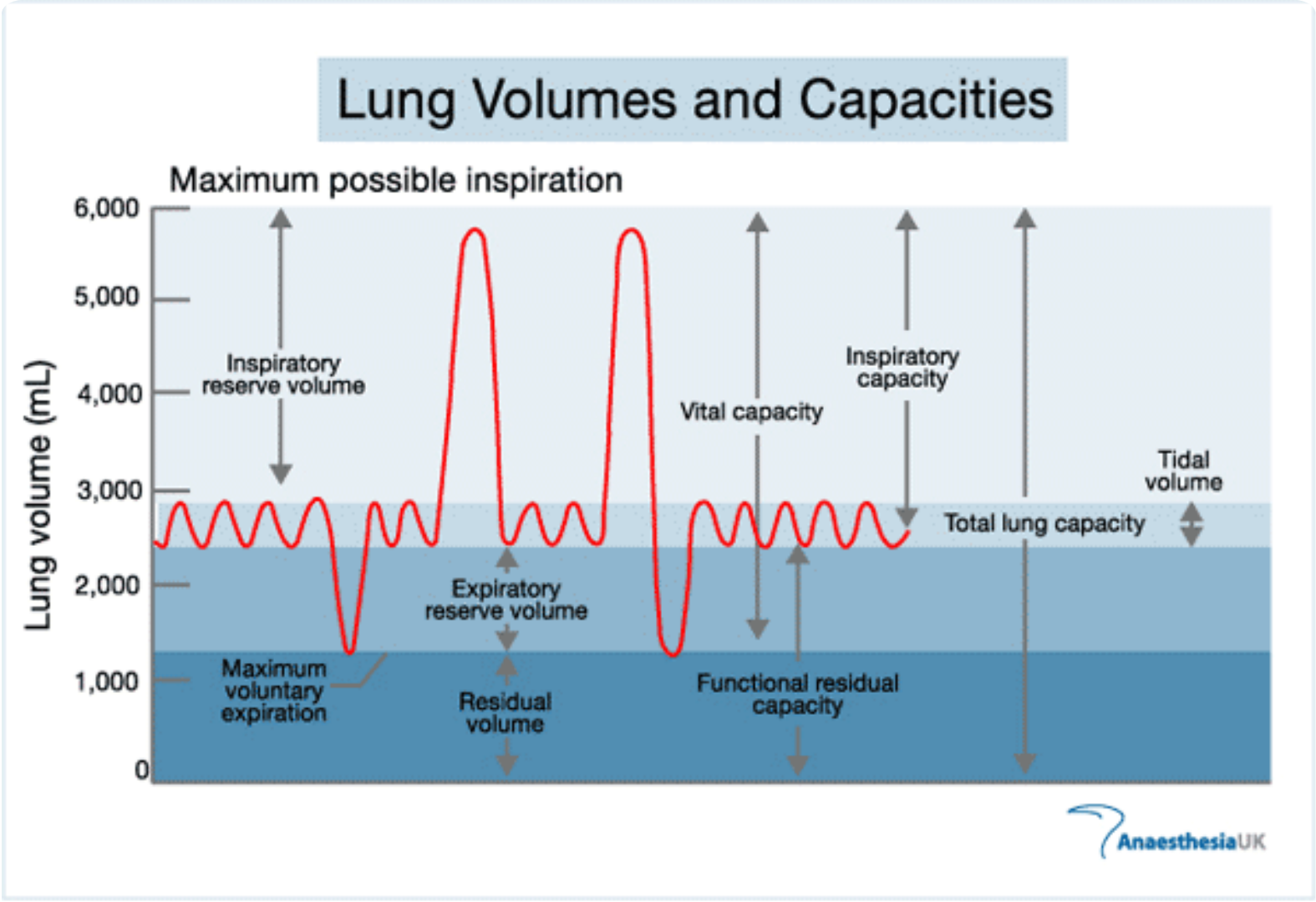
Volume of air in alveoli that are ventilated but not perfused, therefore gas exchange does not take place.



Lung Volumes:

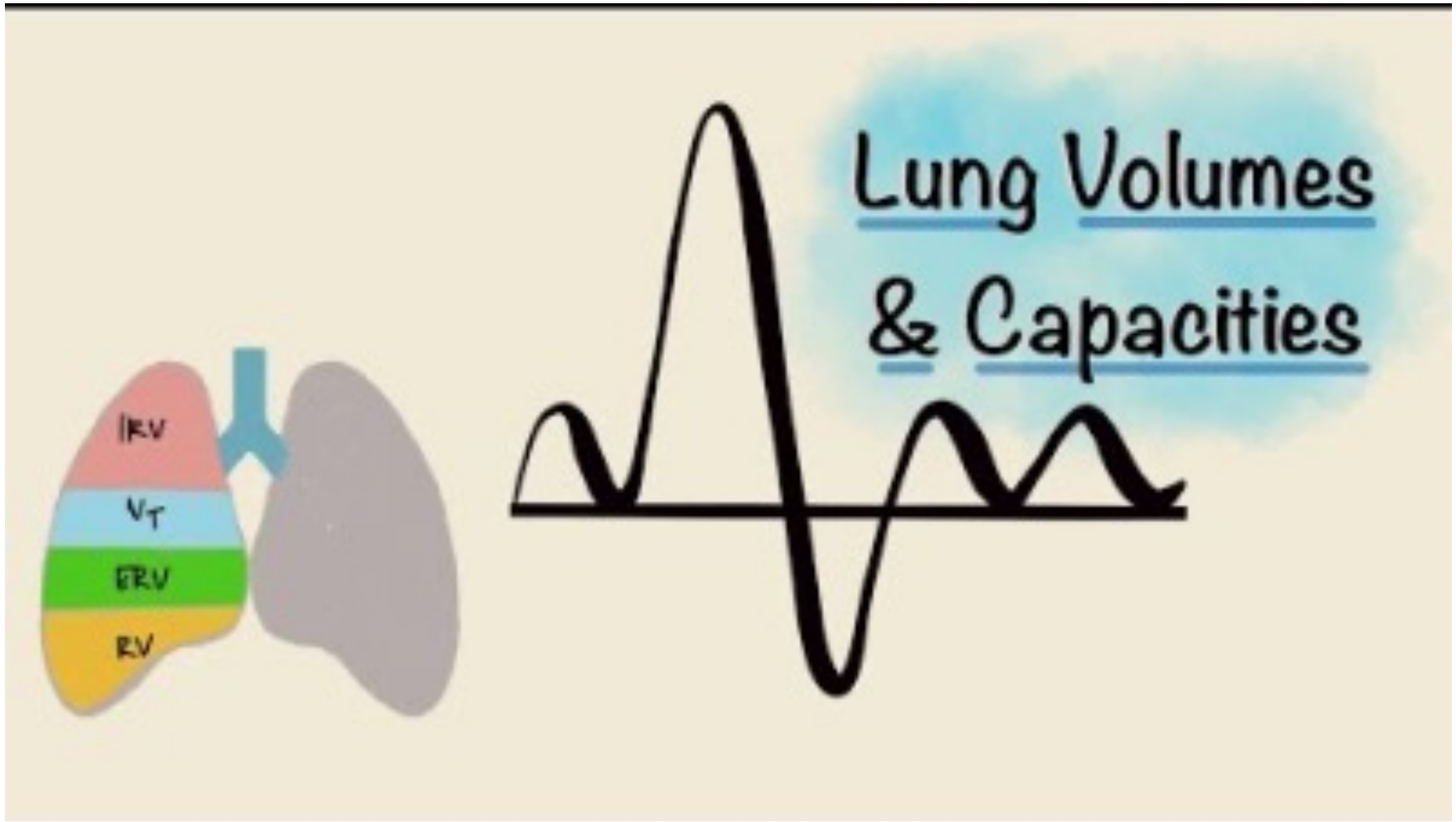
What do the following mean?

- Tidal Volume (TV)
- Inspiratory Reserve Volume (IRV)
- Expiratory Reserve Volume (ERV)
- Functional Residual Capacity (FRC)
- Residual Volume (RV)
- Vital Capacity (VC)
- Total Lung Capacity (TLC)



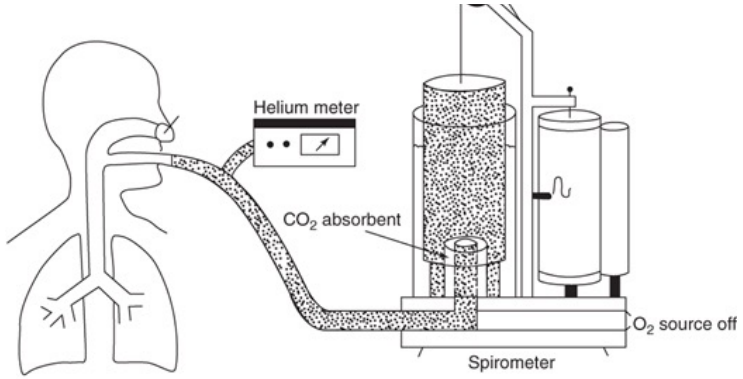
Lung Volumes:

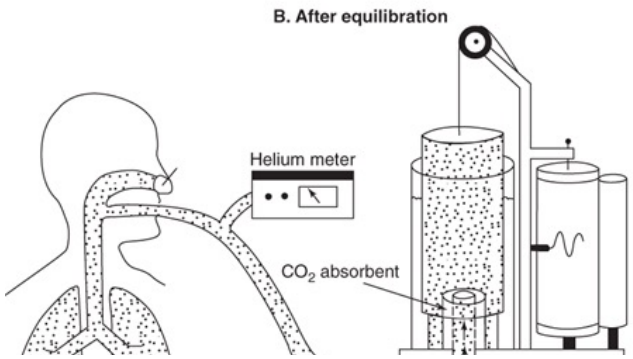
- Tidal Volume (TV) = amount of air inspired during normal, relaxed breathing
- Inspiratory Reserve Volume (IRV) = additional air that can be forcibly inhaled after the inspiration of a normal tidal volume.
- Expiratory Reserve Volume (ERV) = additional air that can be forcibly exhaled after the expiration of a normal tidal volume.
- Functional Residual Capacity (FRC) = is the volume remaining in the lungs after a normal, passive exhalation
- Residual Volume (RV) = volume of air still remaining in the lungs after the expiratory reserve volume is exhaled.
- Vital Capacity (VC) = total amount of air that can be expired after fully inhaling (VC = TV + IRV + ERV = approximately 80 percent TLC)
- Total Lung Capacity (TLC) = maximum amount of air that can fill the lungs (TLC = TV + IRV + ERV + RV)


IRV
V_T
ERV
RV
Lung Volumes
& Capacities

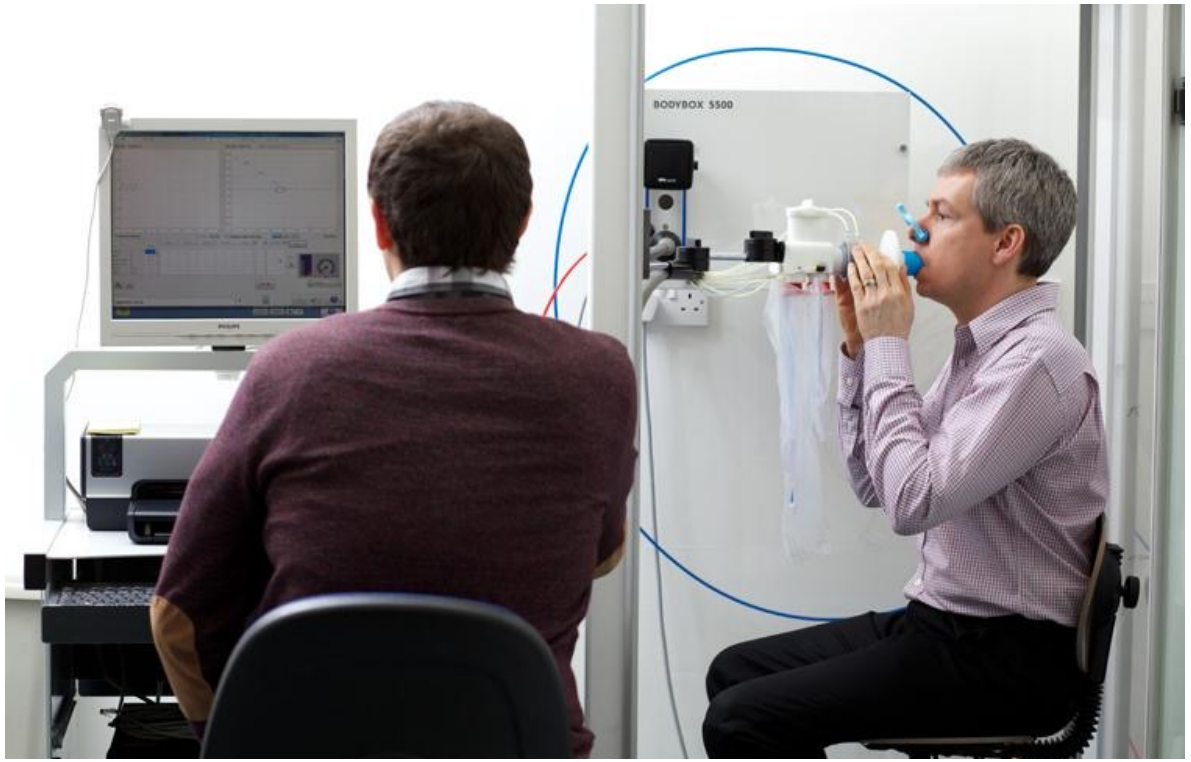
How do we measure lung volumes?

Helium dilution:





Spirometry:



Lung Function Testing - Spirometry:

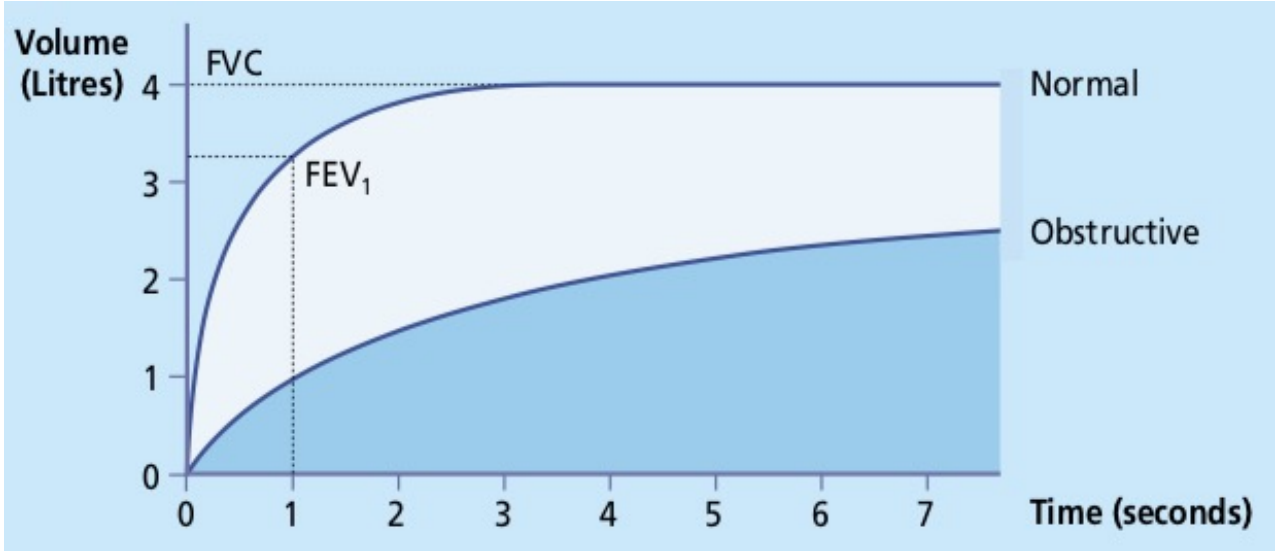
Spirometry gives three important measures:

- FEV1 = the volume of air that the patient is able to exhale in the first second of forced expiration
- FVC = the total volume of air that the patient can forcibly exhale in one breath
- FEV1/FVC = the ratio of FEV1 to FVC expressed as a percentage

- Values of FEV1 and FVC are expressed as a percentage of the predicted normal for a person of the same sex, age and height

Spirometry can also be used to measure:

- VC = slow vital capacity
- FEV1/VC = the ratio of FEV1 to the slow vital capacity



Minute ventilation & alveolar ventilation

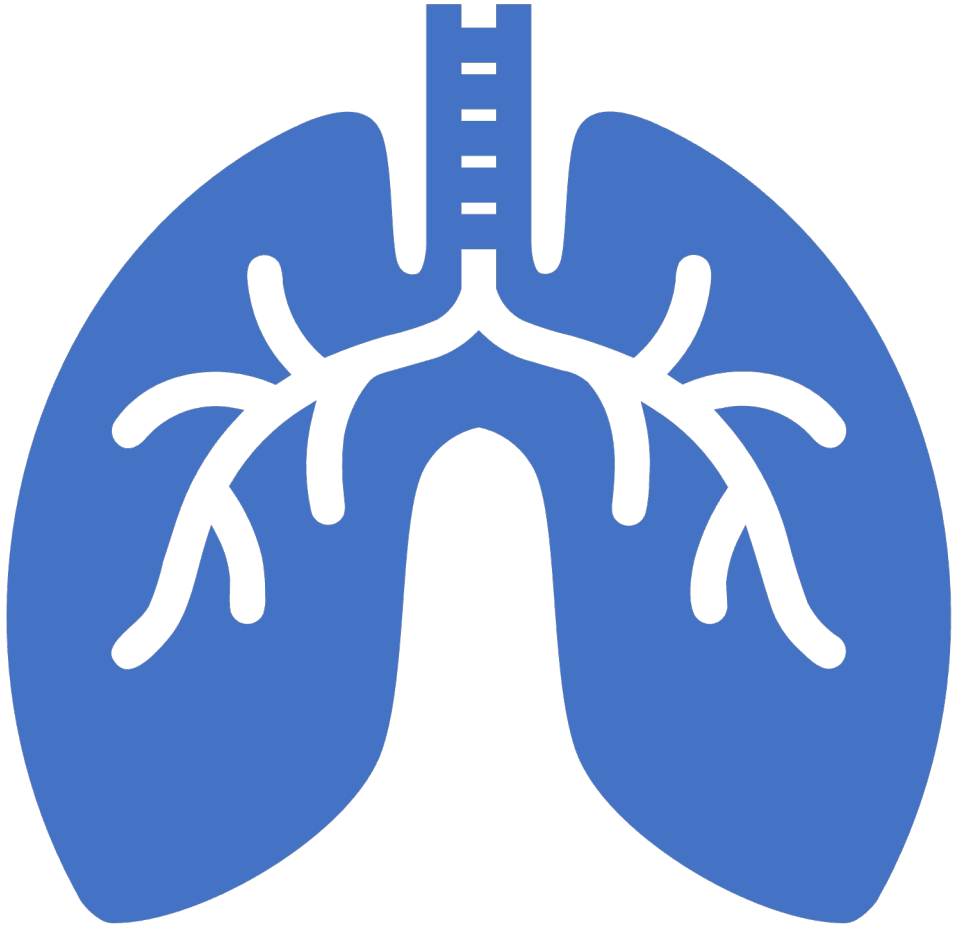
Minute ventilation = the volume of gas inhaled (inhaled minute volume) or exhaled (exhaled minute volume) from a person's lungs per minute.

MV = RR x TV

Alveolar ventilation = the volume of gas entering the alveoli per minute

AV cannot be measured directly – but is determined from TV and physiological dead space

AV= RR(TV-VD) where R is respiratory rate, VT is tidal volume, and VD is dead space volume.



Lung Volumes - Reference slide:

Tidal volume (TV) is the volume of air moved in and out of the respiratory tract (breathed) during each ventilatory cycle.

Inspiratory reserve volume (IRV) is the additional volume of air that can be forcibly inhaled following a normal inspiration. It can be accessed simply by inspiring maximally, to the maximal inspiratory level.

Expiratory reserve volume (ERV) is the additional volume of air that can be forcibly exhaled following a normal expiration. It can be accessed simply by expiring maximally to the maximal expiratory level.

Vital capacity (VC) is the maximal volume of air that can be forcibly exhaled after a maximal inspiration. VC = TV + IRV + ERV.

Residual volume (RV) is that volume of air remaining in the lungs after a maximal expiration. It cannot be expired no matter how vigorous or long the effort. RV = FRC - ERV.

Functional residual capacity (FRC) is the volume of air remaining in the lungs at the end of a normal expiration. FRC = RV + ERV.

Total lung capacity (TLC) is the volume of air in the lungs at the end of a maximal inspiration. TLC = FRC + TV + IRV = VC + RV

Minute volume is the volume of air exhaled per minute.

Maximal breathing capacity (also called "maximal voluntary ventilation") is the maximum volume of air that can be exhaled by voluntary effort in a 15 second interval. This volume is multiplied by 4 and expressed as litres per minute.

Forced expiratory volume 1 (FEV_1) - the volume of air that is forcefully exhaled in one second.

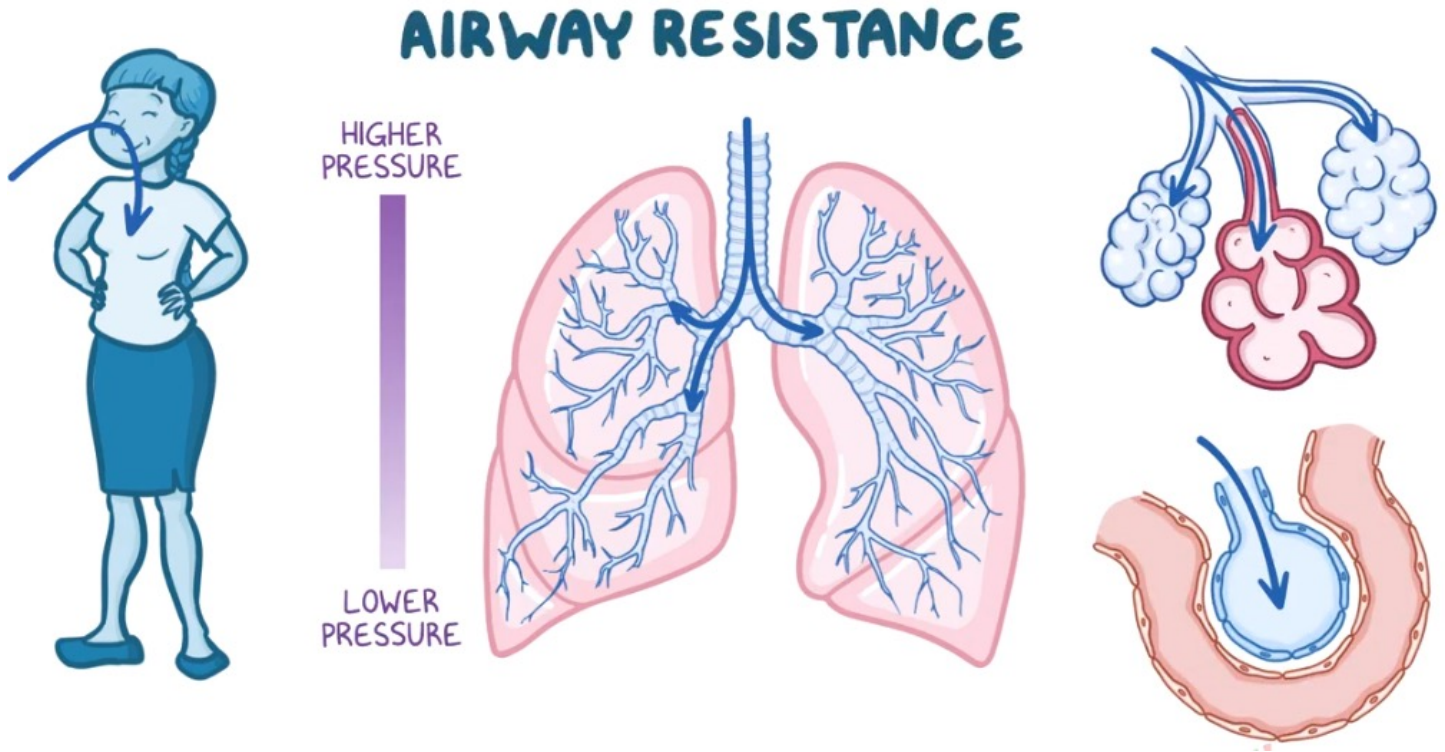
Forced vital capacity (FVC) - the volume of air that can be maximally forcefully exhaled.

Ratio of FEV_1 to FVC (FEV_1/FVC) - expressed as a percentage.

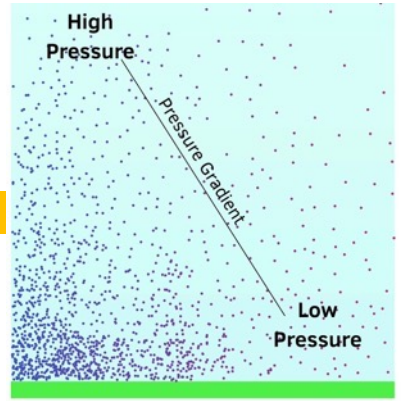
Forced expiratory flow (FEF25 - 75) - the average forced expiratory flow during the mid (25 - 75%) portion of the FVC.

Peak expiratory flow rate (PEFR) - the peak flow rate during expiration.

Airflow – Pressure – Resistance

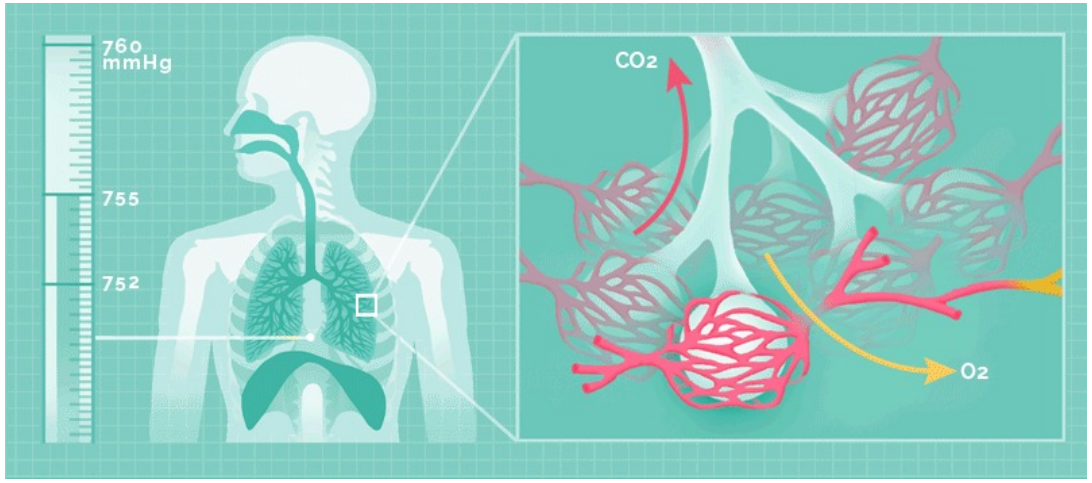


Mechanics of ventilation:



Ventilation requires cyclical movement of air into/out of lungs

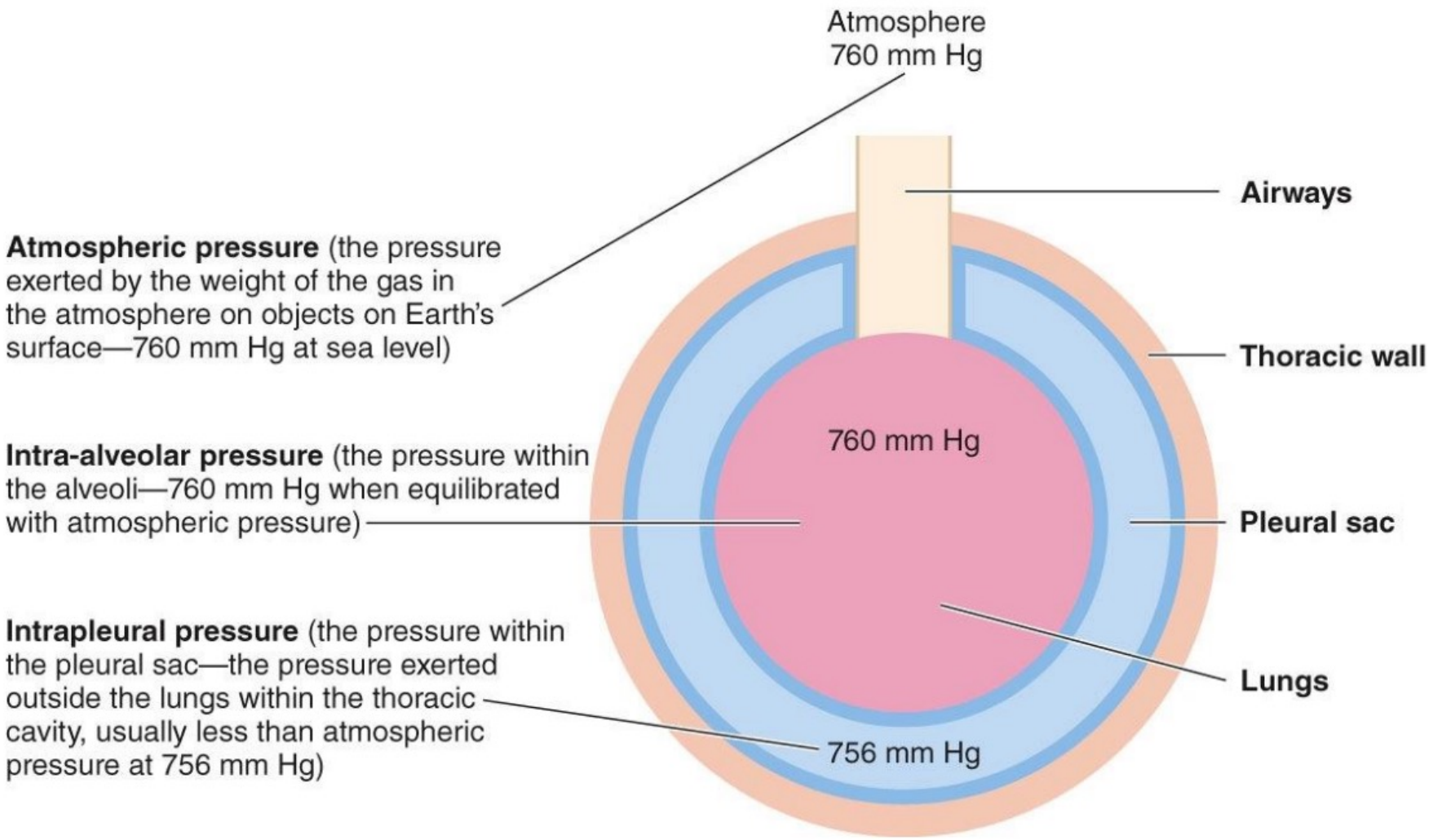
- Pressure gradients exist between alveoli & atmosphere
- Air flows in/out of lungs - alternately reversing pressure gradients



Atmospheric pressure

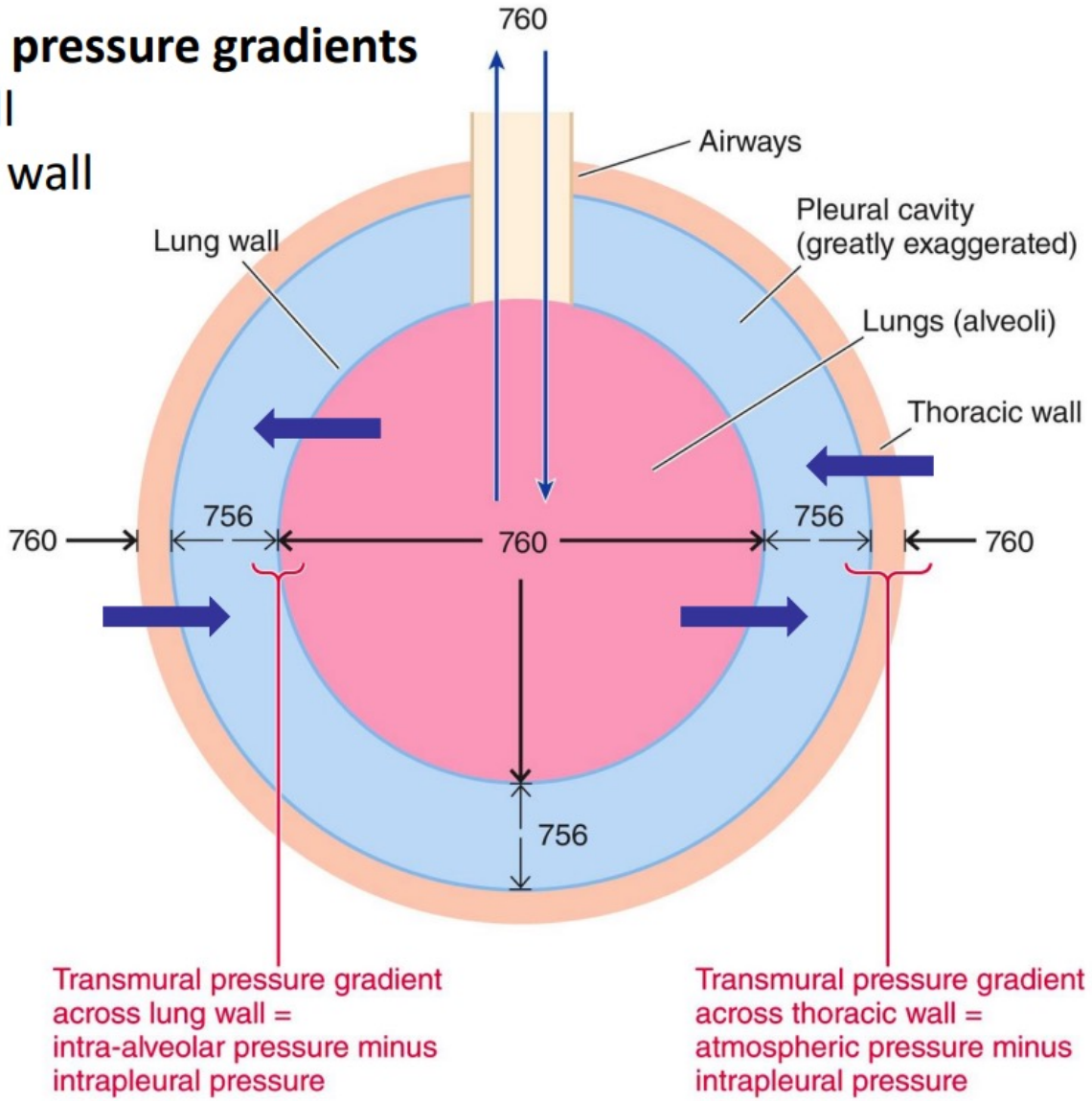
Intra-alveolar pressure aka intrapulmonary

Intrapleural pressure aka intrathoracic


Atmosphere
760 mm Hg
Atmospheric pressure (the pressure exerted by the weight of the gas in the atmosphere on objects on Earth's surface—760 mm Hg at sea level)
Airways
Thoracic wall
760 mm Hg
Intra-alveolar pressure (the pressure within the alveoli—760 mm Hg when equilibrated with atmospheric pressure)
Pleural sac
Intrapleural pressure (the pressure within the pleural sac—the pressure exerted outside the lungs within the thoracic cavity, usually less than atmospheric pressure at 756 mm Hg)
Lungs
756 mm Hg

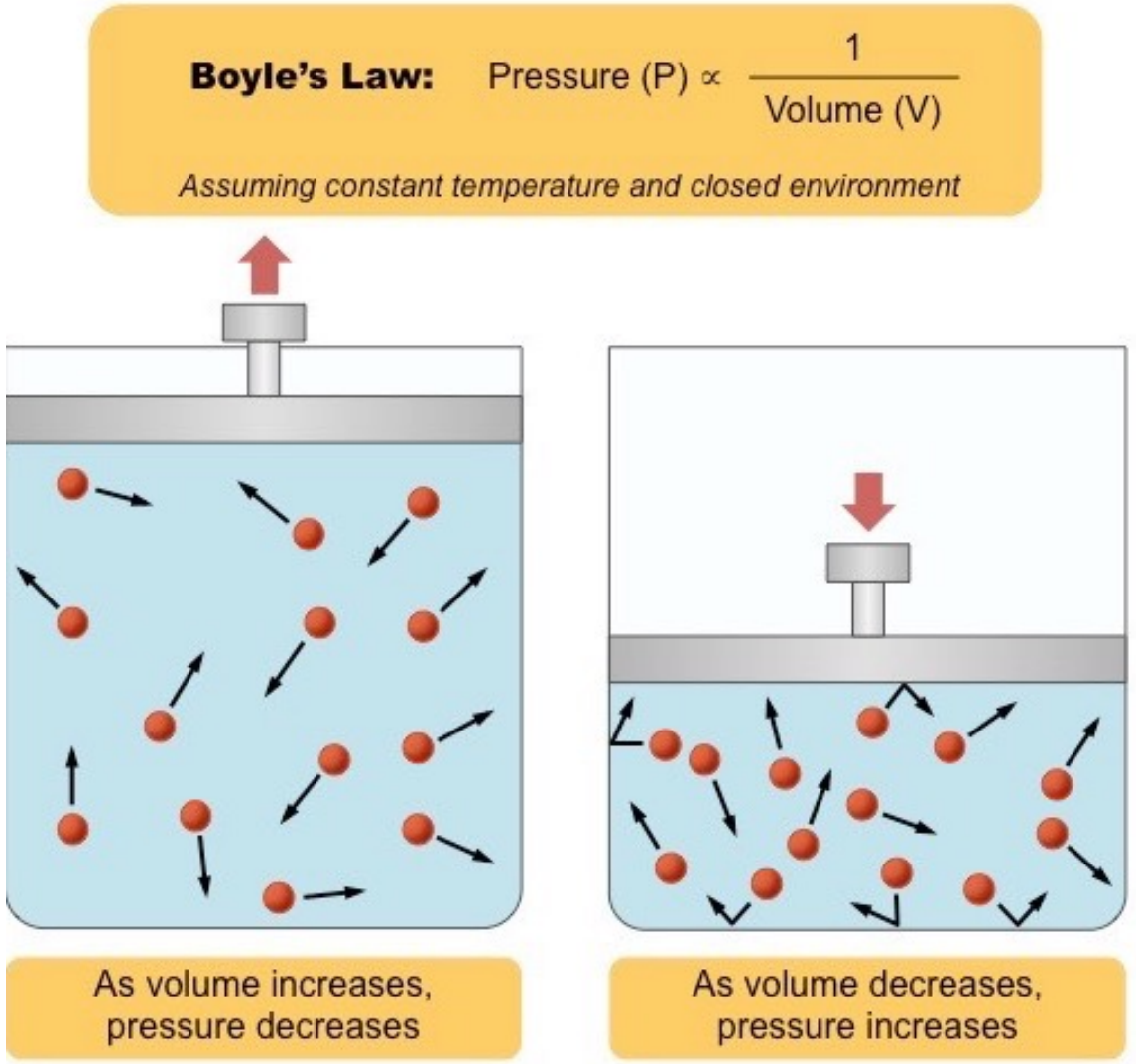
Transmural pressure gradients

- Lung wall
- Thoracic wall



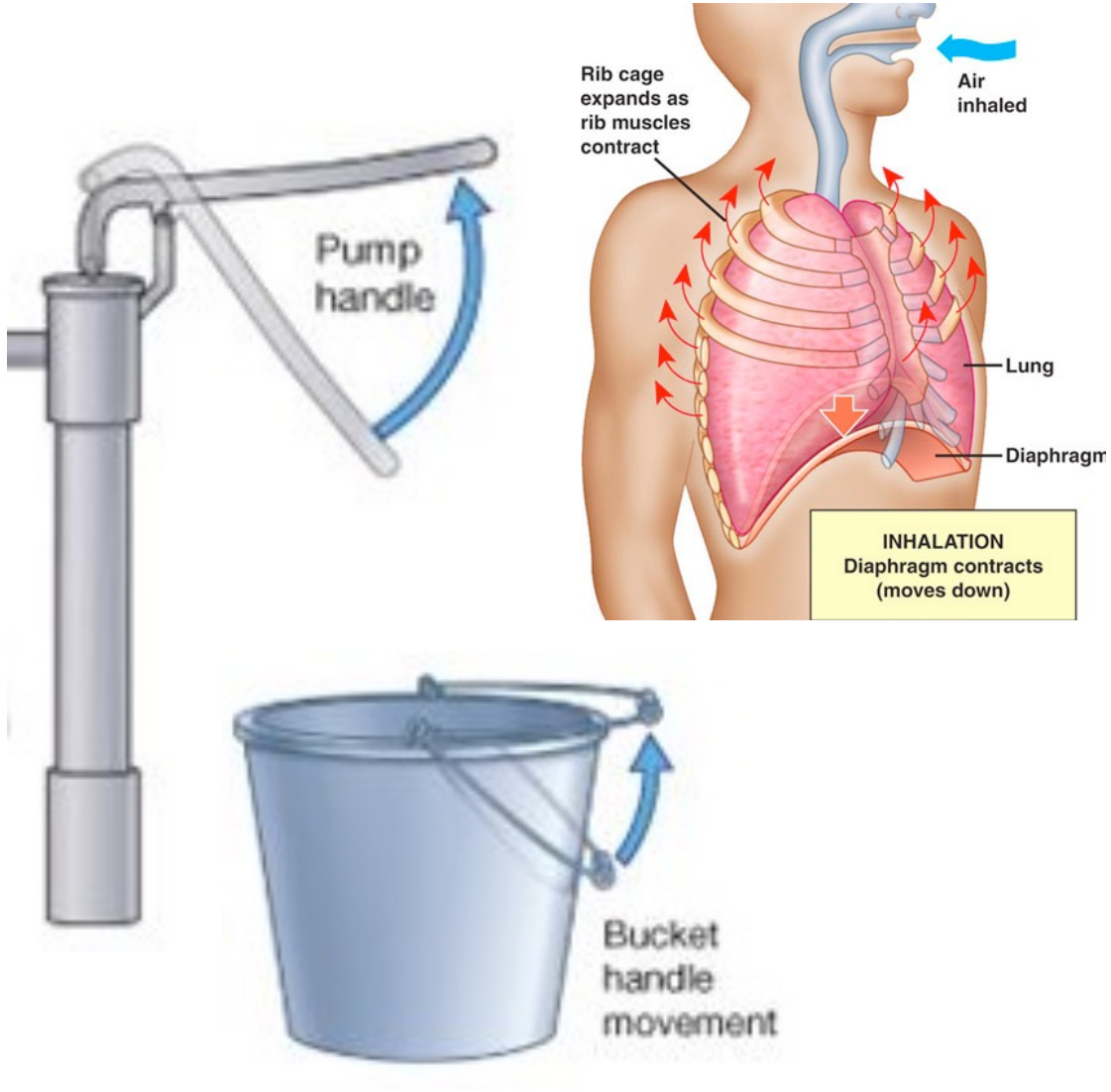
BOYLES LAW:

- Changes in intrapulmonary pressure occur as result of changes in lung volume
  - Pressure of gas is inversely proportional to volume
- Increase in lung volume = decreased intrapulmonary pressure
  - Air goes in
- Decrease in lung volume raises intrapulmonary pressure above atmosphere
  - Air goes out



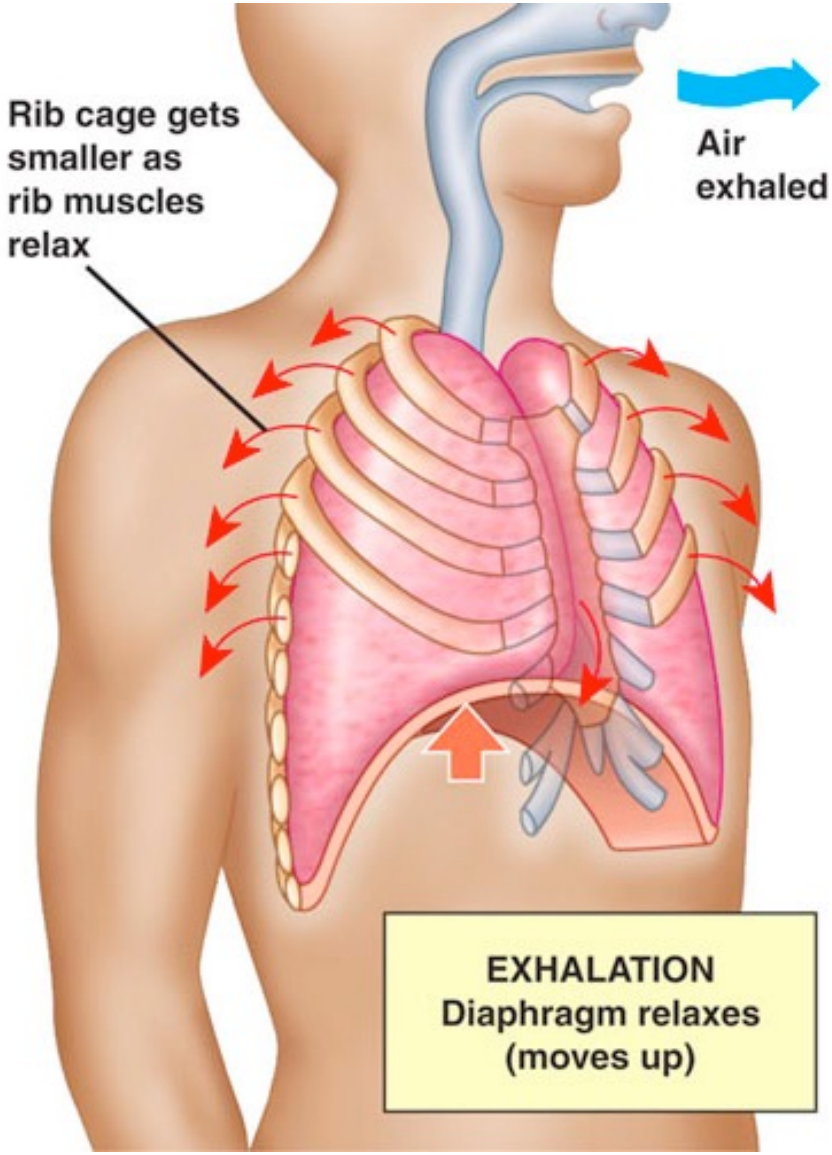
Mechanics of Breathing: Inhalation

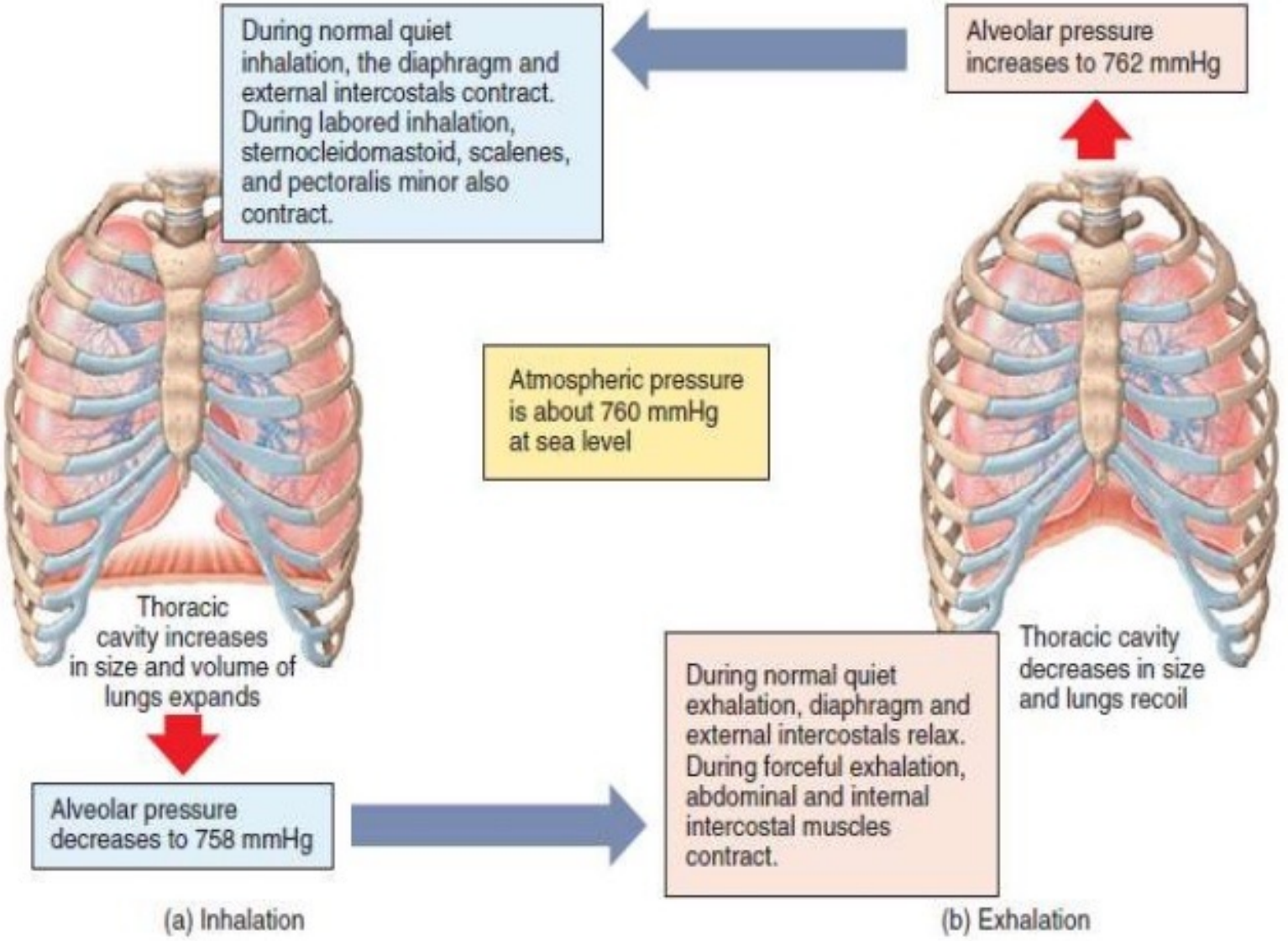
- Diaphragm contracts and flattens
- External intercostal muscles draw the ribs upward and outward
- Increase in thoracic volume
- Decrease in intrapulmonary pressure
- Active process



Mechanics of Breathing: Exhalation

- Diaphragm and external intercostal muscles relax
- Thoracic cavity volume decreases
- Increased intrapulmonary pressure - exceeds the atmospheric pressure
- Air is expelled from the lungs
- Passive process




During normal quiet inhalation, the diaphragm and external intercostals contract. During labored inhalation, sternocleidomastoid, scalenes, and pectoralis minor also contract.
Alveolar pressure increases to 762 mmHg
Atmospheric pressure is about 760 mmHg at sea level
Thoracic cavity increases in size and volume of lungs expands
Alveolar pressure decreases to 758 mmHg
(a) Inhalation
During normal quiet exhalation, diaphragm and external intercostals relax. During forceful exhalation, abdominal and internal intercostal muscles contract.
Thoracic cavity decreases in size and lungs recoil
(b) Exhalation

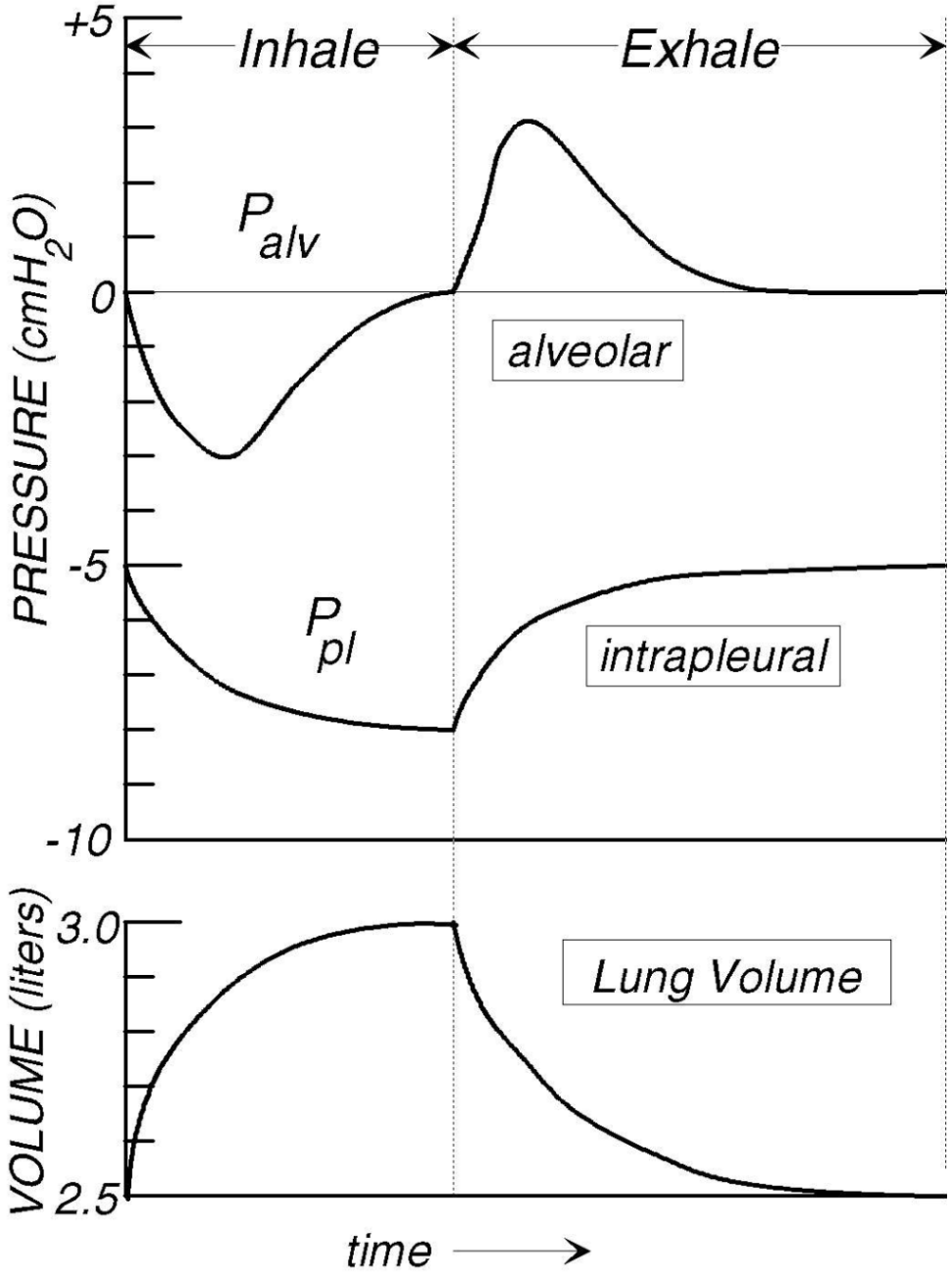
Alveolar pressure

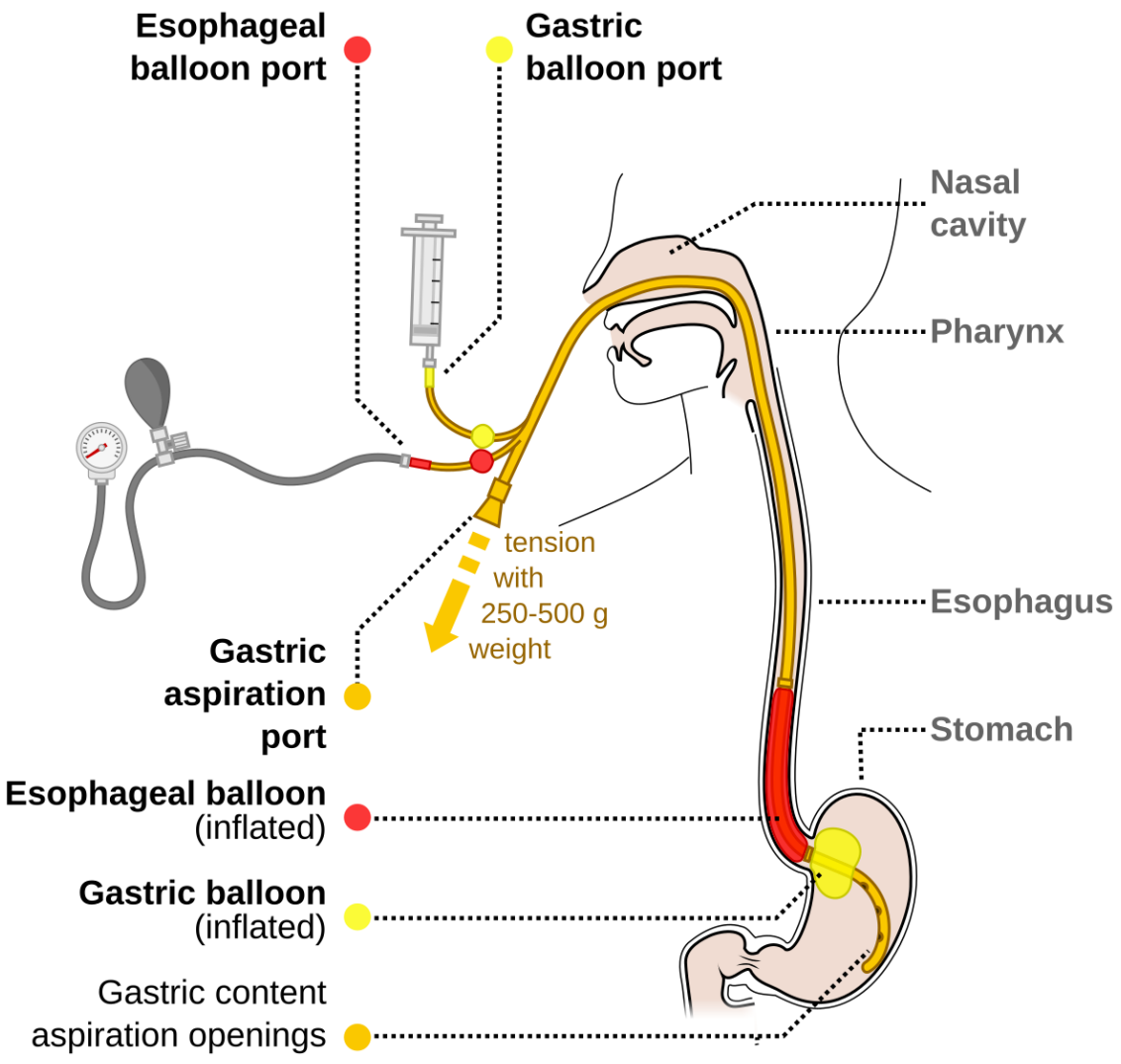
Changes during inhalation & exhalation….

During inhalation, where the alveolar volume increases alveolar pressure decreases (-1cm/h2o) …. Why is this important?

During exhalation, where the alveoli collapse, alveolar pressure increases (+1cm/h2o)… Why is this important?

How does alveolar pressure influence airflow during deep inspiration/expiration?





How can we measure pleural pressure?

- Via the nose through into the oesophagus
- Oesophagus shares pleural space with lungs and heart
- Changes in intra oesophageal pressure mimic closely intra pleural pressure

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Questions & Close

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